The transitive closure is extensive and transitive: A is in its trcl and trcl(A) is transitive

The Open Logic Project contributors

openlogic:set-theory/spine:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both sentences of the proof are verbatim in the span; the staged definition it unfolds is inside the same chapter file.

Editorial note. The label subsetoftrcl names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Finite closure stages cl_n(A). Same unit, the definition preceding the proposition (source lines 450-463): cl_0(A) = A, cl_{n+1}(A) = union of members, trcl(A) = union over n < omega of cl_n(A). Both halves of the proof unfold these definitions directly.

Proposition 0.1. *$A \subseteq \operatorname{trcl} A$ and $\operatorname{trcl} A$ is a transitive set.*

Proof. Evidently $A = \mathrm{cl}_0(A) \subseteq \mathrm{trcl}(A)$. And if $x \in b \in \operatorname{trcl} A$, then $b \in \mathrm{cl}_n(A)$ for some n, so $x \in \mathrm{cl}_{n+1}(A) \subseteq \mathrm{trcl}(A)$. □